EE236: Electronic Devices Lab
Lab 8: Hall Effect Sensor

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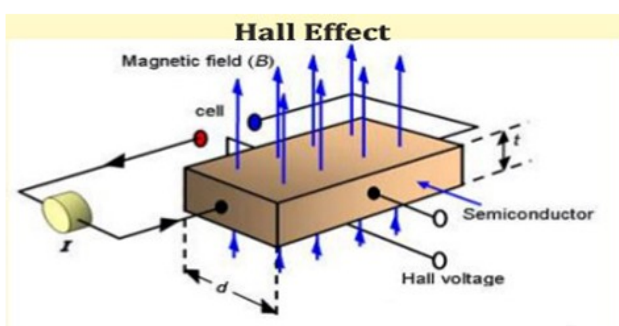
1 Aim of the experiment

- To study the Hall Effect.
- to determine the strength of the given bar magnet and calculate the doping concentration of the hall element in the given sensor.

2 Theory

Hall effect suggests that the voltage generated in a conductor is directly proportional to applied magnetic field and also to the current passing through the conductor. Hall effect sensor uses this phenomenon and generates output voltage in response to applied magnetic flux.



2.1 Bar Magnet strength

The Hall element is a square of side length $a = 0.53\,\text{mm}$, input voltage $V_{\text{in}} = 8\,\text{V}$, and mobility $\mu = 0.8\,\text{m}^2/\text{Vs}$.
The longitudinal electric field is given by

$$E_{\text{long}} = \frac{V_{\text{in}}}{a} \tag{1}$$

The lateral electric field due to the Hall effect is given by

$$E_{\text{lat}} = \frac{|V_{\text{out},0} - V_{\text{out,max}}|}{a} \tag{2}$$

where $V_{\text{out,max}}$ is the maximum V_{out} from the plot in the three magnets case when the magnet is very close to the sensor, and $V_{\text{out},0}$ is the output voltage when no magnetic field is applied.
At equilibrium and assuming a constant magnetic field, the relationship is given by

$$E_{\text{lat}} = 3B \cdot \mu \cdot E_{\text{long}} \tag{3}$$

where B is the magnetic field strength of one bar magnet (in T).

2.2 Doping concentration

Given the thickness of the Hall element $t = 0.053\,\text{mm}$, the Hall coefficient R_H is given by

$$R_H = \frac{V_{\text{out,max}} \cdot t}{I_{\text{in}} \cdot B} \tag{4}$$

The doping concentration N (in cm^{-3}) is given by

$$N = \frac{1}{e \cdot R_H} \tag{5}$$

where $e = 1.6 \times 10^{-19}\,\text{C}$.

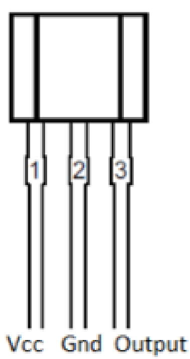
3 Components required

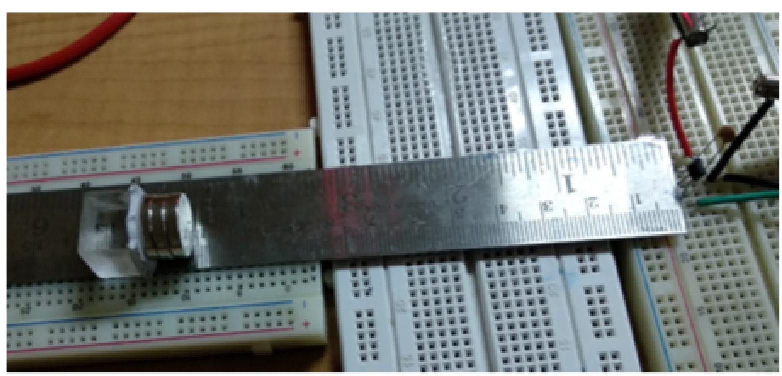
The following components are needed in order to perform the experiment:

- 3-terminal hall sensor
- 3 bar magnets
- Multimeter
- 15cm scale
- Breadboard
- Connecting wires

4 Experiment

4.1 Setup

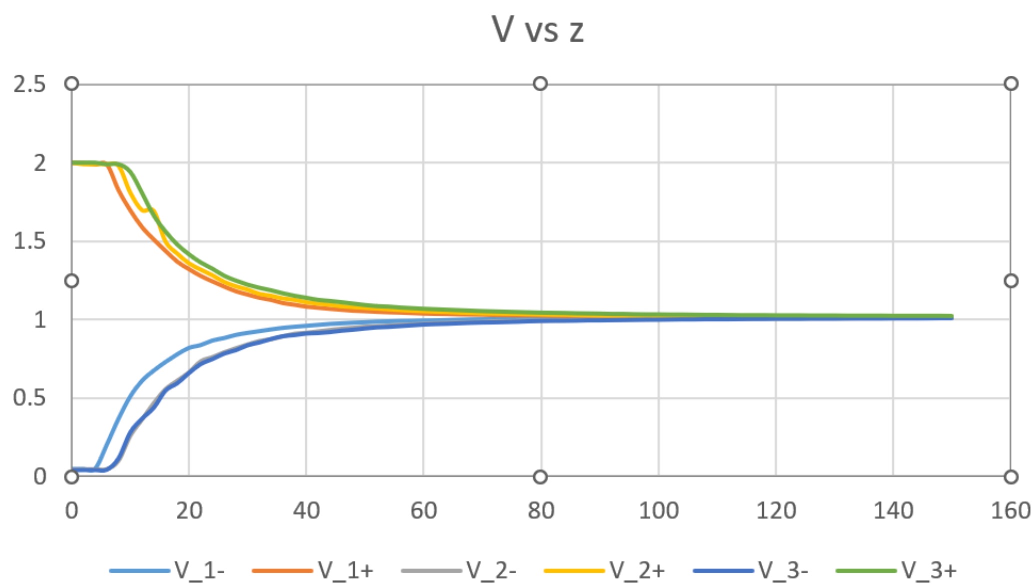




4.2 Readings

z (in mm)	V_{1^-}	V_{1^+}	V_{2^-}	V_{2^+}	V_{3^-}	V_{3^+}
0	0.043	2	0.042	2	0.042	2
2	0.043	2	0.043	1.99	0.043	2
4	0.045	1.99	0.043	1.99	0.043	2
6	0.199	1.99	0.044	1.99	0.045	1.99
8	0.365	1.826	0.099	1.979	0.118	1.99
10	0.507	1.698	0.258	1.809	0.283	1.941
12	0.607	1.591	0.362	1.698	0.37	1.806
14	0.672	1.512	0.467	1.693	0.44	1.662
16	0.727	1.44	0.556	1.499	0.549	1.56
18	0.776	1.371	0.611	1.422	0.596	1.477
20	0.817	1.323	0.664	1.36	0.661	1.414
22	0.835	1.28	0.735	1.32	0.718	1.362
24	0.864	1.245	0.762	1.284	0.75	1.322
26	0.88	1.212	0.793	1.242	0.785	1.277
28	0.9	1.182	0.819	1.214	0.807	1.246
30	0.913	1.161	0.843	1.193	0.838	1.22
32	0.924	1.141	0.865	1.166	0.856	1.2
34	0.935	1.127	0.877	1.154	0.878	1.184
36	0.945	1.107	0.895	1.137	0.896	1.165
38	0.952	1.096	0.912	1.128	0.903	1.149
40	0.958	1.084	0.918	1.113	0.913	1.136
42	0.964	1.077	0.927	1.104	0.916	1.123
44	0.97	1.07	0.936	1.097	0.922	1.116
46	0.974	1.064	0.944	1.087	0.931	1.108
48	0.978	1.058	0.95	1.082	0.937	1.099
50	0.981	1.055	0.957	1.077	0.945	1.09
52	0.985	1.051	0.96	1.073	0.952	1.083
54	0.987	1.048	0.964	1.068	0.955	1.08
56	0.989	1.046	0.969	1.063	0.96	1.074
58	0.99	1.043	0.972	1.059	0.965	1.07
60	0.992	1.04	0.976	1.056	0.97	1.066
62	0.993	1.038	0.98	1.053	0.973	1.063
64	0.995	1.037	0.982	1.051	0.975	1.06
66	0.997	1.035	0.984	1.047	0.978	1.057
68	0.999	1.034	0.986	1.045	0.981	1.054
70	1	1.033	0.988	1.042	0.983	1.051
72	1.001	1.031	0.992	1.041	0.984	1.049
74	1.003	1.03	0.992	1.04	0.986	1.047

76	1.003	1.029	0.994	1.039	0.988	1.045
78	1.004	1.029	0.995	1.037	0.991	1.043
80	1.005	1.028	0.996	1.036	0.993	1.041
82	1.005	1.027	0.997	1.034	0.994	1.04
84	1.006	1.026	0.998	1.034	0.994	1.039
86	1.007	1.026	1	1.032	0.995	1.037
88	1.007	1.025	1.001	1.031	0.997	1.036
90	1.008	1.024	1.002	1.03	0.997	1.035
92	1.008	1.024	1.002	1.029	0.998	1.033
94	1.009	1.023	1.003	1.029	0.999	1.033
96	1.009	1.023	1.004	1.028	1	1.031
98	1.009	1.022	1.004	1.027	1.001	1.03
100	1.009	1.022	1.005	1.027	1.001	1.03
102	1.01	1.022	1.005	1.026	1.002	1.029
104	1.01	1.022	1.006	1.026	1.003	1.029
106	1.01	1.021	1.006	1.025	1.003	1.028
108	1.011	1.021	1.007	1.024	1.004	1.027
110	1.011	1.021	1.007	1.024	1.004	1.027
112	1.011	1.02	1.007	1.024	1.004	1.026
114	1.011	1.02	1.008	1.024	1.005	1.025
116	1.011	1.02	1.008	1.023	1.006	1.025
118	1.011	1.02	1.009	1.023	1.006	1.025
120	1.012	1.019	1.009	1.022	1.007	1.024
122	1.012	1.019	1.009	1.022	1.007	1.024
124	1.012	1.019	1.009	1.022	1.007	1.024
126	1.012	1.019	1.01	1.022	1.008	1.023
128	1.012	1.019	1.01	1.021	1.008	1.023
130	1.012	1.019	1.01	1.021	1.008	1.023
132	1.013	1.019	1.01	1.021	1.009	1.022
134	1.013	1.018	1.011	1.021	1.009	1.022
136	1.013	1.018	1.011	1.02	1.009	1.022
138	1.013	1.018	1.011	1.02	1.009	1.022
140	1.013	1.018	1.011	1.02	1.01	1.021
142	1.013	1.018	1.011	1.02	1.01	1.021
144	1.014	1.018	1.011	1.02	1.01	1.021
146	1.014	1.018	1.012	1.019	1.01	1.021
148	1.014	1.018	1.012	1.019	1.011	1.021
150	1.014	1.018	1.012	1.019	1.011	1.02



From equation (1),

$$E_{\text{long}} = 15.094 \times 10^3 \tag{6}$$

Similary from (2),

$$E_{\text{lat}} = 1.849 \times 10^3 \tag{7}$$

Therefore,

$$B = \frac{E_{\text{lat}}}{3\mu E_{\text{long}}} = 0.051 \text{ T} \tag{8}$$

Using (4) and I_{in} as 5.57 mA,

$$R_H = 0.373\ m^3/C \tag{9}$$

Therefore,

$$N = 1.675 \times 10^{25}\ cm^3 \tag{10}$$